Q1.

Two pupils heated some copper carbonate in a crucible. They recorded the mass of the crucible and contents before and after heating.



(a) The word equation for this reaction is:

copper carbonate → copper oxide + carbon dioxide

(i) What mass of carbon dioxide is given off in this reaction?
Give the unit.

.....

1 mark

(ii) What is the name of this type of chemical reaction?
Tick the correct box.

combustion ☐

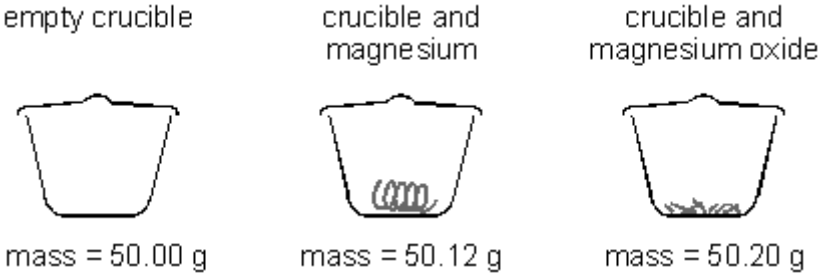
oxidation ☐

reduction ☐

thermal decomposition ☐

1 mark

(b) The pupils then heated some magnesium in another crucible. They worked carefully and did not lose any of the magnesium oxide which formed.
They recorded the mass of the crucible and contents before and after heating.



(i) Write a word equation for the reaction.

.....

1 mark

(ii) Why does the mass of the contents of the crucible increase in this reaction?

.....

.....

1 mark

(iii) What is this type of chemical reaction called?

.....

1 mark

Maximum 5 marks

Q1.

(a) (i) 0.44 g

the unit is required for the mark

1 (L5)

(ii) thermal decomposition ✓

if more than one box is ticked, award no mark

1 (L6)

(b) (i) magnesium + oxygen → magnesium oxide

accept '2Mg + O_2 → 2MgO'

1 (L6)

(ii) magnesium **or** the contents have gained oxygen

accept 'gained oxygen'
accept 'the oxygen has mass'
*do **not** accept 'it had gained gas' **or** 'an oxide is formed'*
***or** 'magnesium reacted with the oxygen'*

1 (L5)

(iii) any **one** from

- oxidation

  *accept 'oxidisation' **or** 'redox'*

- combustion

  accept 'burning'

1 (L6)

[5]